In [5]:
```
# Import necessary packages and Qiskit tools & services
import numpy as np
from qiskit_aer import AerSimulator
from qiskit_ibm_runtime import QiskitRuntimeService
from qiskit_ibm_runtime import Options, Session, Batch, SamplerV2 as Sampler, EstimatorV2 as Estimator
from qiskit.circuit import QuantumCircuit
from qiskit.circuit.random import random_circuit
from qiskit.transpiler.preset_passmanagers import generate_preset_pass_manager
from qiskit.visualization import plot_histogram
from qiskit import __version__
print(__version__)
```

```
1.4.2
```

Initialization

In [123]:
```
# (1) start from |000>
qc_base = QuantumCircuit(3)
```

Randomize

In [124]:
```
# (2) random single- and two-qubit operations
# here we use Qiskit's random_circuit for convenience
rand_qc = random_circuit(num_qubits=3, depth=4, measure=False)
qc_base.compose(rand_qc, inplace=True)
```

Case A: Toffoli (CCX)

In [161]:
```
from qiskit import QuantumCircuit
from qiskit.circuit.random import random_circuit


qc_A = qc_base.copy()
qc_A.ccx(0, 1, 2)  # in-place append of CCX

# draw the full qc_A
qc_A.draw(output="mpl")


# 2. Extract the last instruction tuple (instr, qargs, cargs)
instr, qargs, cargs = qc_A.data[-1]

# 3. Compute the qubit-indices in qc_A
#    qargs is a list of Qubit objects from qc_A.qubits,
#    so we can find their integer positions like this:
qubit_indices = [qc_A.qubits.index(q) for q in qargs]

# 4. Build a new 3-qubit circuit and append *exactly* that CCX
qc_only_ccx = QuantumCircuit(3)
qc_only_ccx.append(instr, qubit_indices, [])

# 5. Draw the minimal CCX circuit
qc_only_ccx.draw(output="mpl")
```

```
C:\Users\sklen\AppData\Local\Temp\ipykernel_38388\255246313.py:13: DeprecationWarning: Treating CircuitInstruction as an iterable is deprecated legacy behavior since Qiskit 1.2, and will be removed in Qiskit 2.0. Instead, use the "operation", "qubits" and "clbits" named attributes.
  instr, qargs, cargs = qc_A.data[-1]
```

Out[161]:

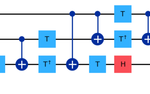
Case B: Toffoli decomposition

In [221]:
```
def apply_toff_decomp(qc, ctrl1, ctrl2, tgt):
    qc.h(tgt)
    qc.cx(ctrl2, tgt)
    qc.tdg(tgt)
    qc.cx(ctrl1, tgt)
    qc.t(tgt)
    qc.cx(ctrl2, tgt)
    qc.tdg(tgt)
    qc.cx(ctrl1, tgt)
    qc.t(ctrl2)
    qc.t(tgt)
    qc.h(tgt)
    qc.cx(ctrl1, ctrl2)
    qc.t(ctrl1)
    qc.tdg(ctrl2)
    qc.cx(ctrl1, ctrl2)

# Build a 3-qubit circuit containing only the decomposition:
qc_toff_decomp = QuantumCircuit(3)
apply_toff_decomp(qc_toff_decomp, 0, 1, 2)

# Draw just the Toffoli-decomposition:
qc_toff_decomp.draw(output="mpl")
```

Out[221]:



Comparison of Density Matrices

In [118]:
```
from qiskit_ibm_runtime import QiskitRuntimeService
from qiskit.transpiler import PassManager

service = QiskitRuntimeService()
backend = service.least_busy(
    operational=True, min_num_qubits=qc_decomp.num_qubits, simulator=False
)
print(backend)
```

```
<IBMBackend'ibm_sherbrooke'>
```

Density Matrix (CCX)

In [119]:
```
from qiskit_experiments.library import StateTomography

qstexp1 = StateTomography(qc_A)
qstdata1 = qstexp1.run(backend, seed_simulation = 100).block_for_results()

# Print total results
for result in qstdata1.analysis_results():
    print(result)
```

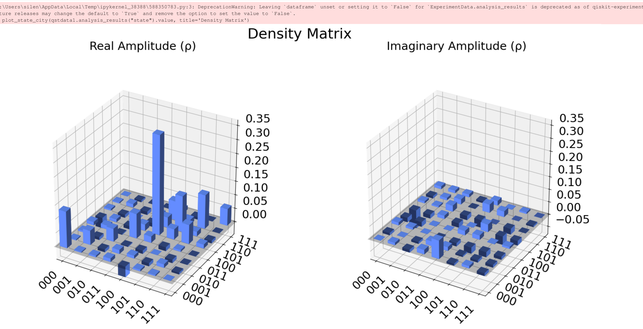
```
AnalysisResult
- name: state
- value: DensityMatrix([[ 1.33199056e-01+0.j        ,  3.48578559e-03+0.00941298j,
                         5.69893296e-03-0.01113553j, 1.23209316e-02+0.01749976j,
                        -3.89743381e-02+0.07231988j, 5.73700469e-03-0.00771181j,
                         6.09535567e-02-0.00939941j, 1.22070324e-04-0.01111984j],
                       [ 3.48578559e-03-0.00941298j, 5.19409180e-02+0.j        ,
                        -1.04573566e-02-0.00797526j, 1.45602756e-02+0.01844619j,
                         1.04980449e-02-0.01529945j, -2.13457576e-02-0.01563691j,
                         1.22070324e-04+0.01111984j, 4.09535567e-02-0.00939941j],
                       [ 5.69893296e-03+0.01113553j, -1.04573566e-02-0.00797526j,
                         4.75322234e-02+0.j        , 2.96396079e-03-0.00092203j,
                        -1.13613237e-02-0.01355566j, 0.00000000e+00-0.02429399j,
                        -1.22082496e-02-0.02006466j, -1.70898438e-03-0.01143392j],
                       [ 1.23209316e-02-0.01749976j, 1.45602756e-02-0.01844619j,
                        -2.96396079e-03+0.00092203j, 9.00539022e-02+0.j        ,
                         2.3072991e-02+0.01733398j, -1.41805013e-02+0.01721191j,
                         6.46972656e-03+0.00138346j, -6.84950087e-04+0.00741914j],
                       [-3.89743381e-02-0.07231988j, 1.04980449e-02-0.01529945j,
                        -1.13613237e-02+0.01355566j, 2.30712891e-02-0.01733398j,
                         3.6058891e-01+0.j        , 3.79096137e-02+0.20032765j,
                         7.02650282e-02-0.03323025j, 1.45626872e-02-0.01802572j],
                       [ 5.73700469e-03+0.00771181j, -1.13457576e-02+0.01563691j,
                         0.00000000e+00+0.02429399j, -1.41805013e-02-0.01721191j,
                         3.79096137e-02-0.20032765j, 1.31218096e-01+0.j        ,
                         1.64622526e-02-0.02785575j, -9.73171658e-03+0.00047472j],
                       [ 6.09535567e-02-0.00939941j, 1.23934514e-02+0.00092302j,
                        -1.22082496e-02+0.02006466j, 6.46972656e-03-0.00138346j,
                         7.02650282e-02+0.03323025j, 1.64622526e-02+0.02785575j,
                         1.24047834e-01+0.j        , -7.84675347e-04-0.05588651j],
                       [ 1.22070324e-04+0.01111984j, 4.09535567e-02+0.00939941j,
                        -1.70898438e-03+0.01143392j, -6.84950087e-04-0.00741914j,
                         1.45626872e-02+0.01802572j, -9.73171658e-03+0.00047472j,
                        -7.84675347e-04+0.05588651j, 6.14237512e-02+0.j        ],
                       dims=(2, 2, 2))
- quality: unknown
- extra: <9 items>
- device_components: ["Q0", "Q1", "Q2"]
- verified: False
AnalysisResult
- name: state_fidelity
- value: 0.37396633696345383
- quality: unknown
- extra: <9 items>
- device_components: ["Q0", "Q1", "Q2"]
- verified: False
AnalysisResult
- name: positive
- value: True
- quality: unknown
- extra: <9 items>
- device_components: ["Q0", "Q1", "Q2"]
- verified: False
```

```
C:\Users\sklen\AppData\Local\Temp\ipykernel_38388\361462133.py:7: DeprecationWarning: Leaving `dataframe` unset or setting it to `False` for `ExperimentData.analysis_results` is deprecated as of qiskit-experiments 0.9.0. F
uture releases may change the default to `True` and remove the option to set the value to `False`.
  for result in qstdata1.analysis_results():
```

In [120]:
```
# Visualize the density matrix above
from qiskit.visualization import plot_state_city
plot_state_city(qstdata1.analysis_results("state").value, title="Density Matrix")
```

```
C:\Users\sklen\AppData\Local\Temp\ipykernel_38388\568350783.py:3: DeprecationWarning: Leaving `dataframe` unset or setting it to `False` for `ExperimentData.analysis_results` is deprecated as of qiskit-experiments 0.9.0. F
uture releases may change the default to `True` and remove the option to set the value to `False`.
  plot_state_city(qstdata1.analysis_results("state").value, title="Density Matrix")
```

Out[120]:



Density Matrix (Toffoli decomposition)

In [123]:
```
qc_toff_decomp_new = qc_base.copy()
apply_toff_decomp(qc_toff_decomp_new, 0, 1, 2)

from qiskit_experiments.library import StateTomography

qstexp2 = StateTomography(qc_toff_decomp_new)
qstdata2 = qstexp2.run(backend, seed_simulation = 100).block_for_results()

# Print total results
for result in qstdata2.analysis_results():
    print(result)
```

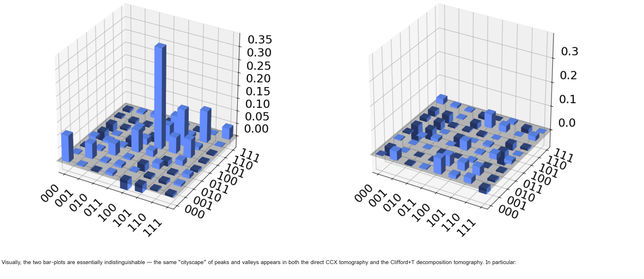
```
AnalysisResult
- name: state
- value: DensityMatrix([[ 0.13413531 +0.j        ,  0.00273302+0.03739421j,
                        -0.00417074+0.00409615j, 0.01017235+0.00649901j,
                        -0.0375434 +0.04631131j, -0.02941895+0.01413981j,
                        -0.05553154-0.02298855j, -0.00994626-0.02303025j],
                       [ 0.00273302-0.03739421j, 0.15960755+0.j        ,
                         0.00634766-0.00356421j, 0.01064464-0.01145128j,
                         0.02231934+0.04449453j, -0.08800239+0.03286404j,
                        -0.01943701+0.03509521j, 0.10742594-0.00834541j],
                       [-0.00417074+0.00409615j, 0.00634766+0.00356421j,
                         0.2477377 +0.j        , 0.10645362-0.01188151j,
                        -0.02862549-0.00056966j, 0.02069092+0.00579834j,
                        -0.00690375-0.00302483j, 0.01416016+0.00423813j],
                       [ 0.01017235-0.00649901j, 0.01064464+0.01145128j,
                         0.10645362+0.01188151j, 0.08717177+0.j        ,
                         0.02301025+0.03901463j, -0.02032471+0.00528971j,
                         0.02026367+0.02244694j, -0.00368084 +0.01381207j],
                       [-0.0375434 -0.04631131j, 0.02231934-0.04449453j,
                        -0.02862549+0.00056966j, 0.03301025-0.03901463j,
                         0.391588  +0.j        , 0.04360541+0.00770312j,
                         0.01341185-0.01780770j, -0.00394076+0.00650124j],
                       [-0.02941895+0.01413981j, -0.08800239-0.03286404j,
                         0.02069092-0.00579834j, -0.02032471+0.00528971j,
                         0.04360541-0.00770312j, 0.14923284+0.j        ,
                         0.01245117-0.02697754j, -0.00494385-0.00653136j],
                       [-0.05553154+0.02298855j, -0.01943701+0.03509521j,
                        -0.00690375+0.00302483j, 0.02026367-0.02244694j,
                         0.01341185+0.01780770j, 0.01245117+0.02697754j,
                         0.12667448+0.j        , 0.00187853+0.01875814j],
                       [-0.00994626+0.02303025j, 0.10742594+0.00834541j,
                         0.01416016-0.00423813j, -0.00368084 -0.01381207j,
                        -0.00394076-0.00650124j, -0.00494385+0.00653136j,
                         0.00187853-0.01875814j, 0.04371474+0.j        ]],
                       dims=(2, 2, 2))
- quality: unknown
- extra: <9 items>
- device_components: ["Q0", "Q1", "Q2"]
- verified: False
AnalysisResult
- name: state_fidelity
- value: 0.42130594756058188
- quality: unknown
- extra: <9 items>
- device_components: ["Q0", "Q1", "Q2"]
- verified: False
AnalysisResult
- name: positive
- value: True
- quality: unknown
- extra: <9 items>
- device_components: ["Q0", "Q1", "Q2"]
- verified: False
```

```
C:\Users\sklen\AppData\Local\Temp\ipykernel_38388\1477092926.py:15: DeprecationWarning: Leaving `dataframe` unset or setting it to `False` for `ExperimentData.analysis_results` is deprecated as of qiskit-experiments 0.9.0.
Future releases may change the default to `True` and remove the option to set the value to `False`.
  for result in qstdata2.analysis_results():
```

In [124]:
```
# Visualize the density matrix above
from qiskit.visualization import plot_state_city
plot_state_city(qstdata2.analysis_results("state").value, title="Density Matrix")
```

```
C:\Users\sklen\AppData\Local\Temp\ipykernel_38388\2029398328.py:3: DeprecationWarning: Leaving `dataframe` unset or setting it to `False` for `ExperimentData.analysis_results` is deprecated as of qiskit-experiments 0.9.0.
Future releases may change the default to `True` and remove the option to set the value to `False`.
  plot_state_city(qstdata2.analysis_results("state").value, title="Density Matrix")
```

Out[124]:



Visually, the two bar-plots are essentially indistinguishable — the same "cityscape" of peaks and valleys appears in both the direct CCX tomography and the Clifford+T decomposition tomography. In particular:

```
Diagonal elements (populations)
Both matrices show the same dominant populations (the tallest real bars) on exactly the same basis states (e.g. that big spike at "100" in your random state).

Off-diagonal coherences
The pattern of smaller real coherences and small imaginary parts is the same in both — you can see the same positive and negative bars in both plots, merely shifted by a few hundredths in height.

Imaginary parts
In both plots the imaginary bars hover around zero with very similar positive/negative patterns and magnitudes.
```

Because the native CCX and its Clifford+T decomposition implement the same three-qubit unitary, their ideal output states are identical. The tiny discrepancies are simply statistical "noise" from finite-shot tomography or device noise, not a failure of the decomposition.